Contest Problem Set 12228

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1. $212 + 21 - 12 =$ _____.
2. The units digit of 942478 is _____.
3. $376 + 245 =$ _____.
4. $422 - 243 =$ _____.
5. The remainder of $166 \div 4$ is _____.
6. $9 + 12 \times 9 =$ _____.
7. $32 + 32 + 48 + 49 =$ _____.
8. The quotient of 119 divided by 7 is _____.
9. $30 \times 19 =$ _____.
10. (estimate) $2409 + 1314 + 2847 =$ _____.
11. $19^2 =$ _____.
12. $616 - 224 - 225 =$ _____.
13. The remainder of $174 \div 9$ is _____.
14. $82 \times 11 =$ _____.
15. $155 + 93 + 62 =$ _____.
16. $1.82 \div 0.013 =$ _____.
17. $35 \times 38 =$ _____.
18. *MMXXII* in Arabic numerals is _____.
19. $18 \times 19 \times 5 =$ _____.
20. (estimate) $249 \times 601 =$ _____.
21. $15 \times 42 \div 30 =$ _____.
22. The greater of $\frac{2}{17}$ and $\frac{3}{26}$ is _____ (fraction).
23. $585 \div 13 =$ _____.
24. The GCD of 65 and 39 is _____.
25. $28^2 =$ _____.
26. $42 + 35 + 28 + 21 + 14 + 7 =$ _____.
27. If 1 ounce is equal to $\frac{1}{16}$ of a pound, then 256 ounces is equal to _____ pounds.
28. $\frac{1}{6} + \frac{1}{10} + \frac{1}{15} =$ _____ (fraction).
29. $99 \times 61 =$ _____.
30. (estimate) $27 \times 303 \times 11 =$ _____.
31. 2520 minutes is _____ hours.
32. The perimeter of a rectangle with width 8 and area 24 is _____.
33. $115^2 =$ _____.
34. 15% of 40 is _____.
35. $41^2 =$ _____.
36. $82 \times 88 =$ _____.
37. The number of odd whole numbers between 100 and 222 is _____.
38. $\frac{24}{25} =$ _____ %.
39. The LCM of 65 and 39 is _____.
40. (estimate) $393852 \div 129 =$ _____.

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41. $77 \times 37 =$ _____.

42. The seventeenth term in the arithmetic sequence $12, 25, 38, \ldots$ is _____.

43. $101 \times 213 =$ _____.

44. The remainder of $1454 \div 11$ is _____.

45. $18 \times 13 + 34 \times 9 =$ _____.

46. The perimeter of an isosceles triangle with sides of length 3 and 8 is _____.

47. $61^2 - 38^2 =$ _____.

48. $2^3 \times 5^4 =$ _____.

49. 1331_4 in base 10 is _____.

50. (estimate) $8333 \times 48 =$ _____.

51. The area of a right triangle with a hypotenuse of length 29 and a leg of length 21 is _____.

52. $105 \times 117 =$ _____.

53. $26\frac{2}{3}\%$ of 30 is _____.

54. $\sqrt[3]{140608} =$ _____.

55. The median of the list $14, 3, 8, 22, 19, 18$ is ____.

56. $375 \times 72 =$ _____.

57. The measure of an interior angle in a regular polygon with 20 sides is _____ °.

58. $96 \times 76 =$ _____.

59. 10000111_2 in base 16 is _____ $_{16}$.

60. (estimate) $857142 \times 3 =$ _____.

61. $0.\overline{074} =$ _____ (fraction).

62. The number of different outcomes showing exactly 3 heads when flipping 5 fair coins is ____.

63. $414 \times 111 =$ _____.

64. 94 in base 3 is _____ $_3$.

65. If $u = 15$, then $u^2 + 32u + 256 =$ _____.

66. $5\frac{2}{3} \times 13\frac{2}{3} =$ _____ (mixed number).

67. $\frac{5}{12} =$ _____ % (mixed number).

68. The area of an equilateral triangle with side length $6\sqrt[4]{3}$ is _____.

69. $16 \times 24 \times 23 =$ _____.

70. (estimate) $\sqrt{11^5} =$ _____.

71. The number of positive whole number divisors of 102 is _____.

72. $1023 \times 1017 =$ _____.

73. $9^{3.5} =$ _____.

74. If $8^{x-2} = 125$, then $2^{2x+1} =$ _____.

75. The number of proper subsets of the set $\{J, U, S, T, W, I, N\}$ is _____.

76. $52 \times 31 \times 75 =$ _____.

77. The reciprocal of 0.0625 is _____.

78. The sum of the terms of the infinite geometric sequence $8, -\frac{8}{3}, \frac{8}{9}, \ldots$ is _____.

79. $\sqrt{2} \times \sqrt{54} \times \sqrt{3} =$ _____.

80. (estimate) The surface area of a right circular cylinder with radius 7 and height 10 is _____.

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1.	A	B	C	D	E	11.	A	B	C	D	E	21.	A	B	C	D	E
2.	A	B	C	D	E	12.	A	B	C	D	E	22.	A	B	C	D	E
3.	A	B	C	D	E	13.	A	B	C	D	E	23.	A	B	C	D	E
4.	A	B	C	D	E	14.	A	B	C	D	E	24.	A	B	C	D	E
5.	A	B	C	D	E	15.	A	B	C	D	E	25.	A	B	C	D	E
6.	A	B	C	D	E	16.	A	B	C	D	E	26.	A	B	C	D	E
7.	A	B	C	D	E	17.	A	B	C	D	E	27.	A	B	C	D	E
8.	A	B	C	D	E	18.	A	B	C	D	E	28.	A	B	C	D	E
9.	A	B	C	D	E	19.	A	B	C	D	E	29.	A	B	C	D	E
10.	A	B	C	D	E	20.	A	B	C	D	E	30.	A	B	C	D	E

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1. What is the tens digit of the product of $2 \times 3 \times 5 \times 7 \times 9 \times 11$?

(A) 7 (B) 9 (C) 1 (D) 0 (E) 5

2. For his summer cookout, Nathan purchased nine packs of hot dogs. There were ten hot dogs in each pack. He also purchased eleven packs of hot dog buns. There were eight hot dog buns in each pack. How many more hot dogs than hot dog buns did Nathan purchase?

(A) 4 (B) 5 (C) 1 (D) 0 (E) 2

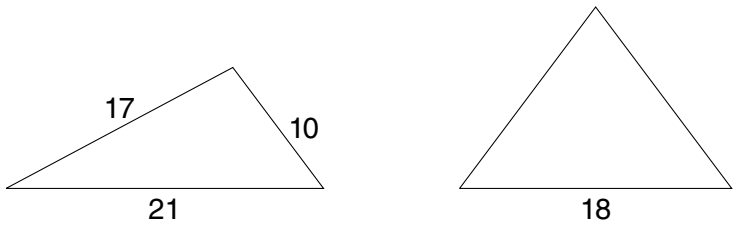
3. Which of the following expressions is odd?

(A) $9 \times 3 + 11$ (B) $4 \times 3 + 6$ (C) $9 \times 1 + 5$ (D) $5 \times 7 + 8$ (E) $8 \times 3 + 4$

4. Quarters are coins worth 25 cents and dimes are coins worth 10 cents. Penny has seven quarters and ten dimes. She wants to purchase some candy that costs 400 cents. How many additional quarters does Penny need to purchase the candy?

(A) 1 (B) 4 (C) 3 (D) 5 (E) 7

5. Two triangles have the same perimeter. One triangle has sides of length 10, 17, and 21. The other triangle has one side of length 18, and the other two sides have equal length. What is the length of one of the two other sides?



(A) 18 (B) 19 (C) 15 (D) 16 (E) 14

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6. Matthew and Mark are learning how to solve a Rubik's Cube. Matthew can scramble and then solve a Rubik's Cube in five minutes, and Mark can scramble and then solve a Rubik's Cube in six minutes. Last week, Matthew and Mark each spent one and a half hours scrambling and then solving a Rubik's Cube. How many more times than Mark did Matthew solve the Rubik's Cube last week?

(A) 2 (B) 1 (C) 3 (D) 5 (E) 4

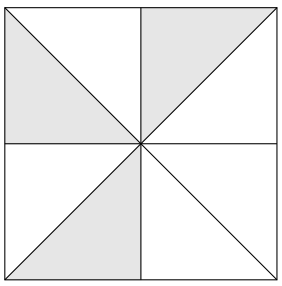
7. How many positive whole numbers less than 100 are evenly divisible by 3 and evenly divisible by 5, but not evenly divisible by 9?

(A) 4 (B) 3 (C) 2 (D) 6 (E) 5

8. Danielle wrote down all of the positive whole numbers less than 100 that were 3 more than multiples of 4. The first number Danielle wrote down was 3, the second was 7, and the third was 11. Danielle made sure that every two-digit number she wrote was greater than the last number she wrote. How many numbers did Danielle write before she wrote the number 67?

(A) 15 (B) 19 (C) 17 (D) 18 (E) 16

9. The square below has an area of 48 and is divided into a number of triangles by two diagonals and two segments connecting the middle points of opposite sides. What is the total area of the shaded triangles?



(A) 15 (B) 20 (C) 21 (D) 16 (E) 18

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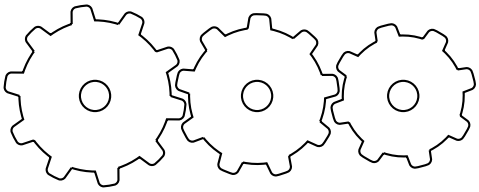
10. Sebastian is preparing a sports drink for his soccer team. The recipe for the sports drink requires adding $\frac{1}{8}$ cup of the sports drink powder for every 1 quart of water. There are 4 quarts in 1 gallon, and Sebastian wants to make 5 gallons of sports drink for his soccer team. How many cups of sports drink powder does Sebastian need?

(A) 2 (B) $3\frac{1}{2}$ (C) 5 (D) $2\frac{1}{2}$ (E) $1\frac{1}{2}$

11. What is the remainder when $11 \times 22 \times 33 \times 44$ is divided by 9?

(A) 3 (B) 0 (C) 8 (D) 2 (E) 6

12. As illustrated below, a gear with 10 teeth is connected to a gear with 9 teeth. The gear with 9 teeth is connected to a gear with 8 teeth. When the gear with 10 teeth makes 180 full turns, how many full turns will the gear with 8 teeth make?



(A) 250 (B) 225 (C) 200 (D) 144 (E) 240

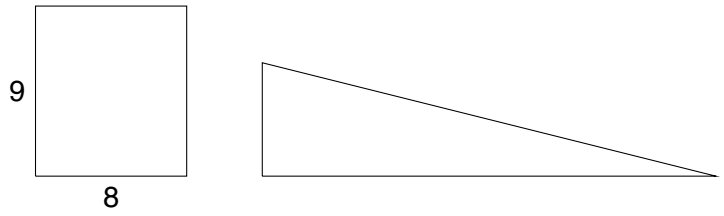
13. The year in which the Continental Army won the Battle of Saratoga, the first major American victory of the Revolutionary War, was a prime number that occurred in the decade of the 1770s. Which of the following was the number of that year?

(A) 1773 (B) 1771 (C) 1779 (D) 1775 (E) 1777

14. Sasha received a book on Monday that he needed to read for a language arts test on Friday. He read $\frac{1}{3}$ of the book on Monday, $\frac{1}{4}$ of the book on Tuesday, and $\frac{1}{5}$ of the book on Wednesday. He read the final 52 pages of the book on Thursday. How many pages long was the book?

(A) 252 (B) 210 (C) 240 (D) 180 (E) 224

15. A rectangle and right triangle have the same area. The rectangle has side lengths of 8 and 9. The length of one leg of the right triangle is four times the length of the other leg of the right triangle. What is the sum of the lengths of the legs of the right triangle?



(A) 26 (B) 24 (C) 28 (D) 30 (E) 32

16. Sylvia was born on May 4th, 2010, which was a Tuesday. Sylvia turned one year old on May 4th, 2011, which was a Wednesday. On what day of the week did Sylvia turn twelve years old?

(A) Tuesday (B) Monday (C) Wednesday (D) Sunday (E) Saturday

17. If $a\heartsuit b = 4ab + 2b - 2a$, then what is the value of $23\heartsuit 22$?

(A) 2022 (B) 2024 (C) 2025 (D) 2021 (E) 2020

18. Jason, Kyla, Lara, Nick, and Mason are classmates that will be sitting around a circular table. Jason does not want to sit next to Kyla, and Mason does not want to sit next to Lara. If a seating arrangement is only considered different if at least one person has someone different sitting on their right, then how many different ways can the five classmates be seated around the circular table?

(A) 10 (B) 9 (C) 6 (D) 8 (E) 12

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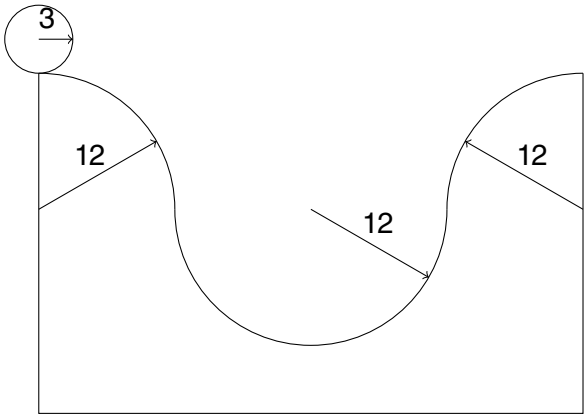
19. A circular wheel with a radius of 3 centimeters rolls, without slipping, along a path that is a quarter-circle of radius 12 centimeters, a semicircle of radius 12 centimeters, and a second quarter-circle of radius 12 centimeters, as illustrated below. How many complete turns does the wheel make when rolling along this path?



(A) 3 (B) 6 (C) 4 (D) 8 (E) 12

20. How many digits are required to write the value of $5^8 \cdot 9^7 \cdot 8^3$?

(A) 13 (B) 15 (C) 12 (D) 16 (E) 14

21. In any triangle, the sum of the lengths of any two sides of the triangle must be greater than the length of the third side. A triangle is a scalene triangle if no two sides of the triangle have the same length. How many different scalene triangles have positive whole number side lengths and perimeters less than 13? Two triangles are considered different only if the lengths of their sides are different.

(A) 4 (B) 3 (C) 2 (D) 6 (E) 5

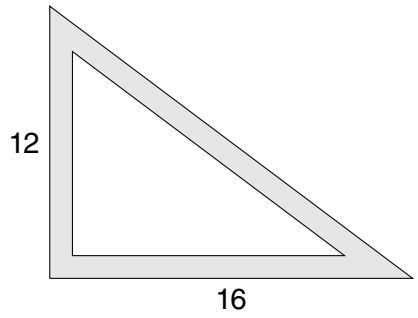
22. When the product of 195 and 273 is written in binary, or base 2, how many more ones than zeros are used?

(A) 5 (B) 8 (C) 4 (D) 7 (E) 6

23. To celebrate her grandchildren coming to visit, a grandmother is going to give a Snickums candy bar, a Dr. Goodbar candy bar, a Four Musketeers candy bar, and a Kat Kit candy bar to her three grandchildren. But to be nice, the grandmother is going to make sure every child receives at least one of the candy bars. How many different ways can the grandmother give the candy bars to her grandchildren?

(A) 48 (B) 24 (C) 72 (D) 36 (E) 18

24. From a piece of paper in the shape of a right triangle with legs of length 16 centimeters and 12 centimeters, a smaller right triangle is cut out and removed. The sides of the smaller right triangle are parallel to the sides of the piece of paper. After the smaller triangle is cut out and removed, the area of the remaining piece of paper is 42. To the nearest whole number, what is the perimeter, in centimeters, of the smaller right triangle?



(A) 40 (B) 38 (C) 36 (D) 39 (E) 37

25. How many odd whole numbers evenly divide 2022^{22}?

(A) 506 (B) 484 (C) 1058 (D) 1012 (E) 529

26. Sean has a small motorboat that he uses to travel upstream and downstream in a river. The motorboat travels at a constant speed of 10 miles per hour in still water. One day when the river current was moving at a constant speed, Sean piloted his motorboat $2\frac{2}{5}$ miles downstream and then immediately turned around and piloted the motorboat back to his starting location. The total time for this trip was 30 minutes. What speed, in miles per hour, was the river current?

(A) 5 (B) 2 (C) 1 (D) 4 (E) 3

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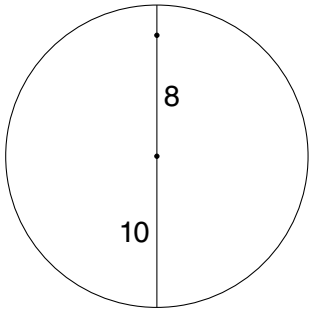
27. How many positive whole numbers are less than one million and have digits that increase in value from left to right? For example, 1489, 267, and 5 are three such numbers, but 1448 is not, since there is no increase from the 4 in the hundreds digit to the 4 in the tens digit.

(A) 451 (B) 511 (C) 465 (D) 484 (E) 475

28. In a circle of radius 10 units, a point is located 8 units from the center of the circle. How many chords include that point and have a length that is a whole number of units? One such chord is the diameter shown below.



(A) 16 (B) 2 (C) 8 (D) 18 (E) ∞

29. Lenny and Benny were investigating the diagonals of various rectangles with perimeters of 12. Lenny told Benny, "If you tell me the length of the diagonal, I can multiply the square of that length by this first number, and then subtract that result from this second number, and the result will always be the area of the rectangle!" What is the product of Lenny's first number and Lenny's second number?

(A) 12 (B) 9 (C) 18 (D) 24 (E) 6

30. If x and y are positive whole numbers such that $y^2 - x^2 = 7y + 6x + 9$, then what is the value of xy?

(A) 192 (B) 144 (C) 216 (D) 96 (E) 256

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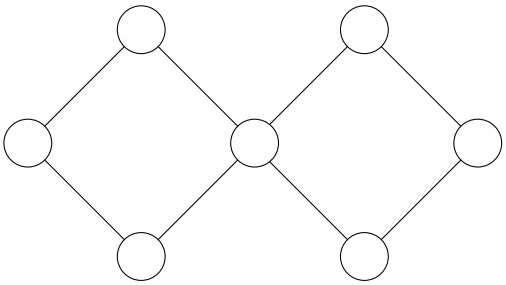
Problems 1 & 2

1. Mackenzie gave some stickers to her friends. She gave Susan seven stickers. She gave Regina two more stickers than she gave Susan. She split the remaining stickers evenly between Kelly, Loraina, and Josephine, so that each received the same number of stickers. Josephine received five fewer stickers than Regina received. How many stickers did Mackenzie give to her friends?

1.

2. The diagram below shows two squares, with circles at the corners of the squares. One circle is shared by both squares. Edgar places each of the numbers 1, 2, 3, 4, 5, 6, and 7 into the circles, so that each number is written in exactly one circle. He then sums the numbers in the four circles at the corners of one square, and also sums the numbers in the four circles at the corners of the other square. Edgar's two sums were the same. What is the greatest possible value of one of Edgar's sums?

2.



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Problems 3 & 4

3. Chris wrote down a three-digit whole number greater than 800 but less than 900. Chris swapped the hundreds and units digits of the number, creating a new three-digit number. He then added that new three-digit number to his original three-digit number, and the sum was 1150. What was Chris' original three-digit number?

3.

4. How many positive whole number divisors of 36036 are multiples of 12?

4.

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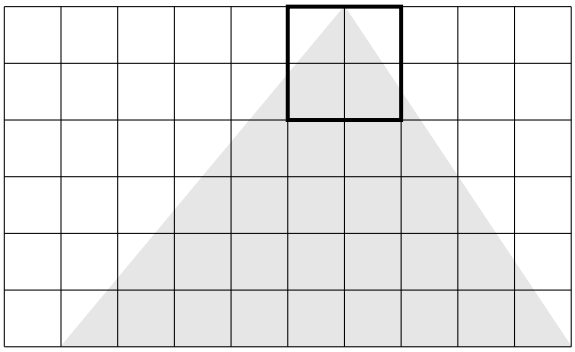
Problems 5 & 6

5. For how many minutes in the five hours between 11:00 and 4:00 do the hour and minute values displayed on a digital clock have digits that sum to 6? For example, one such minute is 3:12, since $3 + 1 + 2 = 6$.

5.

6. A 100×60 rectangle is divided into a number of 10×10 squares as shown. A shaded triangle is drawn with vertices that coincide with corners of the squares, also as shown. A region of four squares is highlighted. What is the area of the portion of the shaded triangle that lies within that region of four squares?

6.



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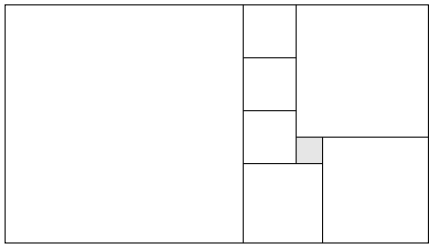
Problems 7 & 8

7. The rectangle below is divided into eight squares as shown. The shaded square has an area of 1. What is the area of the entire rectangle?



7.

8. On El-Al-Gebra Airlines, there is no cost for a bag that weighs no more than a certain weight limit, but there is a extra fee for each pound that a bag weighs above the weight limit. Amy and Bryan each brought a bag on their El-Al-Gebra flight, and the combined weight of their bags was 120 pounds. Each of their bags was over the weight limit. Amy had to pay an extra fee of $20, and Bryan had to pay an extra fee of $10. If Amy and Bryan together had one bag that weighed 120 pounds, the extra fee for being over the weight limit would have been $90. How many pounds is the weight limit for a bag that can be brought on El-Al-Gebra Airlines without an extra fee?

8.

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Team Round 12228

School or Team

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1.	2.	3.	4.	5.
6.	7.	8.	9.	10.

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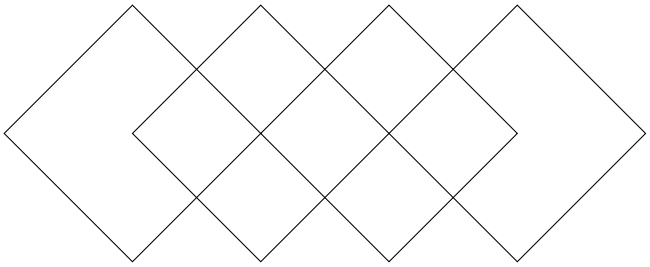
1. Sam wrote down all of the numbers from 1 through 100. How many numbers that Sam wrote had the digit 4, the digit 5, or the digit 6?

2. Travis began watching the final three episodes of *The Book of Boba Fett* at 5:45 PM. The first episode he watched was 50 minutes long, and the second episode was 47 minutes long. If Travis took a break of 5 minutes between one episode and the next, and he finished watching all three episodes at 8:31 PM, then how many minutes long was the third episode?

3. Pierre wrote down three two-digit prime numbers. Pierre used each of the digits 1, 3, 4, 5, 7, and 9 exactly once to write these three numbers. What is the greatest possible sum of Pierre's three prime numbers?

4. Four squares of side length 6 are arranged as shown. The centers and two of the corners of each of the squares are all on the same line, and corners of two squares lie at the center of each of the two middle squares. What is the total area of the region that lies within at least one of the squares?



5. Will has a special deck of 59 cards. Each of the numbers from 1 through 59 appears on exactly one of the cards. Will sorted all of the cards into separate groups, and no group of cards had two cards that summed to a multiple of 10. What is the fewest number of groups of cards that Will could have?

6. In the boxes below, the value of each box is a positive whole number, and the value of each of the three rightmost boxes is the average, or mean, of the values of the two boxes immediately to the left. For example, the value of the box labeled *C* is the average of the values of the boxes labeled *A* and *B*. If the rightmost box has a value of 12, as shown, and the value of the box labeled *D* is not 12, then how many different positive whole number values are possible for the box labeled *A*?

A	B	C	D	12

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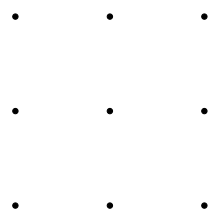
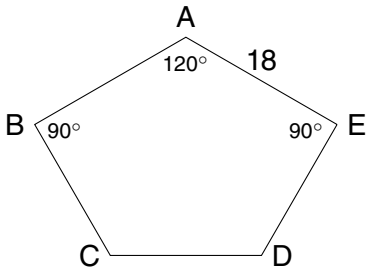
7. The nine dots below are arranged in a 3×3 grid of three evenly-spaced rows and three evenly-spaced columns. How many triangles can be formed by using three dots as vertices?



8. A shipping clerk had to weigh five boxes before shipment. Unfortunately, the shipping clerk only had a scale that could accurately weigh items with weights greater than 80 pounds. The shipping clerk decided to weigh the items in pairs. The shipping clerk weighed every possible pair of boxes, and the scale showed weights, in pounds, of 95, 98, 101, 104, 105, 107, 108, 110, 111, and 117. Using these results, the clerk was able to determine the weight of each of the five boxes. What is the remainder when the product of the five weights, in pounds, of the boxes is divided by 100?

9. Pentagon *ABCDE* is shown below. The interior angle at vertex *A* measures 120°, and the interior angles at vertices *B* and *E* each measure 90°. The measures of the interior angles at vertices *C* and *D* are equal, and the length of side *AE* is 18. Also, the lengths of sides *BC*, *CD*, and *DE* are equal to each other. To the nearest whole number, what is the area of *ABCDE*?



10. Octavius has three fair eight-sided dice, each with sides numbered 1 through 8. Octavius rolls the three dice and then sums the numbers shown. What is the probability that Octavius's sum is divisible by 3? Express your answer as a common fraction.

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Answers 12228

Sprint Round

1. B	11. E	21. B
2. E	12. B	22. B
3. D	13. E	23. D
4. D	14. C	24. C
5. C	15. D	25. E
6. C	16. C	26. B
7. A	17. A	27. C
8. E	18. D	28. A
9. E	19. C	29. B
10. D	20. B	30. B

Target Round

1. 28
2. 17
3. 872
4. 16
5. 24
6. 265
7. 144
8. 48

Team Round

1. 51
2. 59
3. 191
4. 117
5. 6
6. 5
7. 76
8. 76
9. 456
10. $\frac{85}{256}$

Number Sense

1. 221	21. 21	41. 2849	61. $\frac{2}{27}$
2. 8	22. $\frac{2}{17}$	42. 220	62. 10
3. 621	23. 45	43. 21513	63. 45954
4. 179	24. 13	44. 2	64. 10111
5. 2	25. 784	45. 540	65. 961
6. 117	26. 147	46. 19	66. $77\frac{4}{9}$
7. 161	27. 16	47. 2277	67. $41\frac{2}{3}$
8. 17	28. $\frac{1}{3}$	48. 5000	68. 27
9. 570	29. 6039	49. 125	69. 8832
10. $[6242, 6898]$	30. $[85492, 94490]$	50. $[379985, 419983]$	70. $[382, 421]$
11. 361	31. 42	51. 210	71. 8
12. 167	32. 22	52. 12285	72. 1040391
13. 3	33. 13225	53. 8	73. 2187
14. 902	34. 6	54. 52	74. 800
15. 310	35. 1681	55. 16	75. 127
16. 140	36. 7216	56. 27000	76. 120900
17. 1330	37. 61	57. 162	77. 16
18. 2022	38. 96	58. 7296	78. 6
19. 1710	39. 195	59. 87	79. 18
20. $[142167, 157131]$	40. $[2901, 3205]$	60. $[2442855, 2699997]$	80. $[711, 785]$

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Solutions 12228

Sprint Round Solutions

1. The product of 2 and 5 is 10, so the tens digit of the entire product is the units digit of the product of the remaining terms. The units digit of 3×7 is 1, the units digit of 9×11 is 9. Therefore the tens digit of the entire product is $1 \times 9 = \boxed{9}$.

2. Nathan purchased $9 \times 10 = 90$ hot dogs, and $11 \times 8 = 88$ hot dog buns. The number of hot dogs exceeded the number of hot dog buns by $90 - 88 = \boxed{2}$.

3. Each of the expressions is a product followed by a sum. A sum is only odd if one term is even and one term is odd. For $9 \times 3 + 11$, both 9×3 and 11 are odd. For $4 \times 3 + 6$, both terms are even. For $9 \times 1 + 5$, both terms are odd. And for $8 \times 3 + 4$, both terms are even. But 5×7 is odd, and 8 is even, so the expression that is odd is $\boxed{5 \times 7 + 8}$.

4. The total value of Penny's quarters and dimes is $7 \times 25 + 10 \times 10 = 275$ cents. She needs an additional $400 - 275 = 125$ cents. The number of quarters required is $125 \div 25 = \boxed{5}$.

5. The perimeter of the first triangle is $10 + 17 + 21 = 48$. The perimeter of the second triangle is also 48, so the two other sides of the second triangle have a total length of $48 - 18 = 30$. Since these two sides have an equal length, the length of each is $30 \div 2 = \boxed{15}$.

6. A period of 1 and $\frac{1}{2}$ hours is $60 + 30 = 90$ minutes. In 90 minutes, Matthew can scramble and solve a Rubik's Cube $90 \div 5 = 18$ times, while Mark can scramble and solve a Rubik's Cube $90 \div 6 = 15$ times. Matthew's count exceeds Mark's count by $18 - 15 = \boxed{3}$.

7. If a number is divisible by both 3 and 5, then it is divisible by $3 \times 5 = 15$. The multiples of 15 less than 100 are 15, 30, 45, 60, 75, and 90, for a total of 6. However, both 45 and 90 are divisible by 9, so the total count of numbers divisible by both 3 and 5 but not 9 is $6 - 2 = \boxed{4}$.

8. The number 3 is 1 less than 4, or 4×1, the number 7 is 1 less than 8, or 4×2, and the number 11 was 1 less than 12, or 4×3. The number 67 is 1 less than 68, or 4×17. Therefore 67 was the seventeenth number Danielle wrote down, and the count of numbers she wrote prior to 67 is $17 - 1 = \boxed{16}$.

9. The diagonals and segments partitioning opposite sides divide the square into 8 triangles, all of equal area. The area of one of these triangles is $48 \div 8 = 6$, so the shaded area, which is 3 of these triangles, is $6 \times 3 = \boxed{18}$.

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10. The total number of quarts in 5 gallons is $5 \times 4 = 20$. Therefore Sebastian needs $20 \times \frac{1}{8}$ cups of sports drink powder. In cups, this is $\frac{20}{8} = \boxed{2\frac{1}{2}}$.

11. The expression can be rewritten as $(9+2) \times (2 \times 9 + 4) \times (3 \times 9 + 6) \times (4 \times 9 + 8)$. Every term in the expansion of this product is divisible by 9 except $2 \times 4 \times 6 \times 8 = 384$, which, when divided by 9, has a remainder of $\boxed{6}$.

12. When the gear with 10 teeth makes 180 full turns, exactly $180 \cdot 10 = 1800$ teeth of the gear pass the point where the gear is connected to the gear with 9 teeth. The same number of teeth on the gear with 9 teeth must pass that same point, so the gear with 9 teeth will make $\frac{1800}{9} = 200$ turns. Similarly, when the gear with 9 teeth makes 200 turns, the number of turns made by the gear with 8 teeth is $\frac{200 \cdot 9}{8} = \boxed{225}$.

13. The number 1775 has a units digit of 5, and is therefore divisible by 5, so it cannot be prime. The numbers 1773 and 1779 both have digits that sum to multiples of 3, and are therefore divisible by 3, so neither can be prime. Finally, the number 1771 has alternating digit sums that differ by a multiple of 11. That is, the difference between the sum of the units digit and the hundreds digit, 8, and the sum of the tens digit and the thousands digit, also 8, is 0, which is a multiple of 11. Therefore 1771 is divisible by 11. The only remaining possibility, which must be the prime, is $\boxed{1777}$.

14. On Monday, Tuesday, and Wednesday, Sasha read $\frac{1}{3} + \frac{1}{4} + \frac{1}{5} = \frac{47}{60}$ of the book. The remaining portion of the book was $1 - \frac{47}{60} = \frac{13}{60}$ of the book. Since this was 52 pages, the length of the entire book, in pages, was $52 \div \frac{13}{60} = \boxed{240}$.

15. The area of the rectangle is $8 \cdot 9 = 72$. The right triangle has the same area, so the product of the lengths of the legs of the right triangle is $72 \cdot 2 = 144$. If one leg has a length 4 times the length of the other, the square of the length of the shorter leg must be $\frac{144}{4} = 36$, so the shorter leg has length 6. The longer leg has length 24, and the sum of the lengths of the legs is $6 + 24 = \boxed{30}$.

16. A non-leap year is 365 days, or 52 weeks and 1 day. Therefore in a non-leap year, the day of the week for Sylvia's birthday is one day later in the week than it was the previous year. But a leap year is 366 days, so since May 4th is after February 29th, the day of the week for Sylvia's birthday is two days later in the week than it was the previous year. Sylvia turned 12 years old in $2010 + 12 = 2022$. The leap years between 2010 and 2022 are 2012, 2016, and 2020. Therefore Sylvia's birthday will be $12 + 3 = 15$ days later in the week than it was in 2010. This is equivalent to $15 - 7 - 7 = 1$ day later in the week, which is $\boxed{\text{Wednesday}}$.

17. The expression $4ab + 2b - 2a - 1$ can be rewritten as $(2a+1)(2b-1) + 1$. Then $23 \heartsuit 22 = (2 \cdot 23 + 1)(2 \cdot 22 - 1) + 1$, or $47 \cdot 43 + 1 = \boxed{2022}$.

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18. Seat Jason at the table. Then there are 2 possibilities for where Kyla sits, as she must not be in one of the two seats next to Jason. Once Kyla is seated, either Mason or Laura must be in the single seat that is between Kyla and Jason, so there are 2 possibilities for this seat. Finally, once that seat is filled, the remaining two classmates can be seated in the remaining two seats in 2 ways. Therefore the total number of different ways the classmates can be seated is $2 \cdot 2 \cdot 2 = \boxed{8}$.

19. Because two quarter-circles and one semicircle are altogether one entire circle, the total length of the path is equal to the circumference of a circle of radius 12, which is $2 \cdot 12 \cdot \pi = 24\pi$. The circumference of the wheel is $2 \cdot 3 \cdot \pi = 6\pi$, so the number of complete turns by the wheel is $\frac{24\pi}{6\pi} = \boxed{4}$.

20. Because $2^3 = 8$, the number 8^3 can be written as $\left(2^3\right)^3 = 2^9$. Each power of 2 paired with a power of 5 creates a power of 10, and there are 9 powers of 2 and 8 powers of 5, so the entire expression can be rewritten as $10^8 \cdot 2 \cdot 9^7$. The factor of 10^8 will append 8 instances of the digit 0 at the right end of the number. Because $3^2 = 9$, the number 9^7 can be written as $\left(3^2\right)^7$, or $\left(3^7\right)^2$. The value of 3^7 is 2187, and since $21^2 = 441$ while $22^2 = 484$, the value of $\left(3^7\right)^2$ is a little less than $4,840,000$. Therefore the value of $2 \cdot \left(3^7\right)^2$ is a little less than $9,700,000$, and requires 7 digits. When multiplied by 10^8, the total number of digits required is $7 + 8 = \boxed{15}$.

21. If the shortest side of the triangle has length 1, then the other two sides must have equal length, and the triangle is not scalene. If the shortest side of the triangle has length 2, then the other two sides can differ by at most 1, and the possible side lengths are 2, 3, and 4, as well as 2, 4, and 5, for 2 possible triangles. Any greater lengths for the other two sides results in a perimeter not less than 13. If the shortest side of the triangle has length 3, then the other two sides can differ by at most 2, and the possible side lengths are 3, 4, and 5, for 1 possible triangle. Any greater lengths for the other two sides results in a perimeter not less than 13. The total number of possible triangles is $2 + 1 = \boxed{3}$.

22. As the sum of powers of 2, the number 195 is $128 + 64 + 2 + 1$. In binary, this is 11000011_2. As the sum of powers of 2, the number 273 is $256 + 16 + 1$. In binary, this is 100010001. The product of these two binary values is $1100001100000000_2 + 110000110000_2 + 11000011_2 = 1100111111110011_2$. This binary number is 16 digits, with 12 ones and 4 zeros, so the number of ones exceeds the number of zeros by $12 - 4 = \boxed{8}$.

23. There are 4 different candy bars and 3 grandchildren, and since every grandchild will receive at least one candy bar, exactly 1 child will receive 2 candy bars. There are 3 ways to pick this grandchild, and $\binom{4}{2} = 6$ ways to select the candy bars this child receives. There are 2 ways to distribute the remaining two candy bars among the other two grandchildren, so the total number of ways the grandmother can give the candy bars to her grandchildren is $3 \cdot 6 \cdot 2 = \boxed{36}$.

24. Because the smaller right triangle has sides that are parallel to the larger triangle, the smaller triangle is similar to the larger triangle. The area of the larger triangle is $\frac{1}{2} \cdot 12 \cdot 16 = 96$, so the area of the smaller triangle is $96 - 42 = 54$. This is $\frac{54}{96} = \frac{9}{16}$ of the area of the larger triangle. Because the two triangles are similar, the side lengths of the smaller triangle must be $\sqrt{\frac{9}{16}} = \frac{3}{4}$ of the side lengths of the larger triangle. Therefore the legs of the smaller triangle are $\frac{3}{4} \cdot 12 = 9$ and $\frac{3}{4} \cdot 16 = 12$, and by the Pythagorean Theorem, the hypotenuse of the smaller triangle is $\sqrt{9^2 + 12^2} = 15$, for a perimeter of exactly $9 + 12 + 15 = \boxed{36}$.

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25. The prime factorization of 2022 is $2 \cdot 3 \cdot 337$, so the prime factorization of 2022^{22} is $2^{22} \cdot 3^{22} \cdot 337^{22}$. For a divisor to be odd, it can include no powers of 2 in its prime factorization. However, any number that is a product of a power of 3 from 3^0 through 3^{22}, for 23 possible powers of 3, and a power of 337 from 337^0 through 337^{22}, for possible 23 powers of 337, will be an odd divisor of 2022^{22}. Therefore the number of odd divisors is $23 \cdot 23 = \boxed{529}$.

26. Expressed as a decimal, $2\frac{2}{5}$ is 2.4. Time is equal to distance divided by rate. If r is the speed of the river current in miles per hour, the rate Austin travels downstream is $10 + r$, and the time, in hours, for Austin to travel downstream is $\frac{2.4}{10+r}$. Similarly, the time, in hours, for Austin to travel upstream is $\frac{2.4}{10-r}$. Since the entire journey lasted 30 minutes, these times must sum to 0.5 hours, so $\frac{2.4}{10+r} + \frac{2.4}{10-r} = 0.5$, or $\frac{24}{10+r} + \frac{24}{10-r} = 5$. This equation could be solved directly, but consideration of divisors of 24 reveals that $10 - r$ is 8 and $10 + r$ is 12, so the speed of the river current, in miles per hour, is $\boxed{2}$.

27. Since the digits increase from the left to right, the digit 0 cannot appear in any of the numbers. Only the digits 1 through 9 may appear. When given a set of 6 or fewer distinct digits, there is exactly 1 way to order the digits in increasing value from left to right. Therefore there are $\binom{9}{6} = 84$ such numbers that use 6 digits, $\binom{9}{5} = 126$ such numbers that use 5 digits, $\binom{9}{4} = 126$ that use 4, $\binom{9}{3} = 84$ that use 3, $\binom{9}{2} = 36$ that use 2, and $\binom{9}{1} = 9$ that use 1. Altogether, the count of such numbers less than one million is $84 + 126 + 126 + 84 + 36 + 9 = \boxed{465}$.

28. The shortest possible length of the chord occurs in the chord that is perpendicular to the diameter that includes the point. The length of half of this chord is, by the Pythagorean Theorem, $\sqrt{10^2 - 8^2} = 6$, so the length of the chord is 12. As a chord including the point is rotated about the point, its length can be any value between the minimum of 12 and the maximum of the diameter, which is 20. Additionally, each length between 12 and 20 occurs in exactly 2 chords, one by rotating the chord clockwise and the other by rotating the chord counterclockwise. Therefore the total number of chords with a length that is a whole number of units is 1, for the chord of minimal length, plus 1, for the diameter, plus $2 \cdot (19 - 13 + 1) = 14$, which sums to $\boxed{16}$.

29. Let the length of the rectangle be x and the width be y. Then the area is xy, and the square of the diagonal is $x^2 + y^2$. Because the perimeter is 12, $x + y = 6$, and $x^2 + 2xy + y^2 = 36$. Therefore $2xy$, or twice the area, is equal to 36 minus the square of the diagonal. Letting the length of the diagonal be d, this is $2xy = 36 - d^2$, so the area, or xy, is $18 - \frac{1}{2}d^2$. Lenny's first number is $\frac{1}{2}$ and his second number is 18, and their product is $\frac{1}{2} \cdot 18 = \boxed{9}$.

30. The given expression can be rewritten as $y^2 - 7y = x^2 + 6x + 9$, or $y \cdot (y - 7) = (x + 3)^2$. If y and $y - 7$ are factor pairs of a perfect square, then one number is odd and the other is even. The even factor must be the square of an even number v times an odd number s, and the odd factor must be the square of an odd number d times the same odd number s. Therefore $(v^2 \cdot s) - (s \cdot d^2) = \pm 7$, or $s \cdot (v^2 - d^2) = \pm 7$. But 7 is prime and both s and $v^2 - d^2$ are whole numbers. Since v and d are positive, the difference of their squares cannot be 1, so s must be 1, and $v^2 - d^2 = \pm 7$. Furthermore, if $v^2 - d^2 = -7$, but v and d are positive, then $v - d = -1$ and $v + d = 7$. This results in $v = 3$, an impossibility since v is even. Thus $v^2 - d^2 = 7$, and $v = 4$ and $d = 3$. Then $y = 16$ and $y - 7 = 9$, so $(x + 3)^2 = 16 \cdot 9$. Since x is positive, $x + 3 = 12$, and $x = 9$. The value of xy is $9 \cdot 16 = \boxed{144}$.

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Target Round Solutions

1. Susan received 7 stickers, so Regina received $7 + 2 = 9$ stickers. Josephine received 5 fewer sticker than Regina, so Josephine received $9 - 5 = 4$ stickers. Kelly and Loraina also received 4 stickers. The total number of stickers Mackenzie gave to her friends was $7 + 9 + 4 + 4 + 4 = \boxed{28}$.

2. The sum of the numbers 1, 2, 3, 4, 5, 6, and 7 is $1 + 2 + 3 + 4 + 5 + 6 + 7 = 28$. When the number in the circle shared by both squares is removed from this sum, the remaining total must be even, since the sum of the remaining three circles must be the same for both squares. For the greatest possible sum, the number in the circle shared by both squares should be as large as possible. Therefore the number in the circle shared by both squares must be 6, and the sum of the remaining three circles for each square must be $(28 - 6) \div 2 = 11$. The partitioning of the remaining numbers is 7, 3, and 1 for one square, and 5, 4, and 2 for the other square. The greatest value of the sum for one square is $6 + 11 = \boxed{17}$.

3. The hundreds digit of Chris' number was an 8. Since the sum was 1150, and the units digit of the sum is the sum of the hundreds and units digits of the original number, the units digit of Chris' number was $10 - 8 = 2$. The sum of $802 + 208$ is 1010, and $1150 - 1010 = 140$. The tens digit of the original number is counted twice, as it occurs in both the original number and the new number. Therefore the tens digit is $140 \div 20 = 7$, and Chris' original three-digit number was $\boxed{872}$.

4. The number 36036 is $36 \cdot 1001$, and has a prime factorization of $2^2 \cdot 3^2 \cdot 7 \cdot 11 \cdot 13$. For a number to be a multiple of 12, it must have at least one factor of 3 and at least two factors of 2 in its prime factorization. Thus the divisor of 36036 must have a prime factorization of 2 factors of 2, for 1 possibility, and 1 or 2 factors of 3, for 2 possibilities. Additionally, the divisor may have 0 or 1 factors of each of 7, 11, or 13, for 2 possibilities for each. Altogether, the number of possible divisors of 36036 that are also multiples of 12 is $1 \cdot 2 \cdot 2 \cdot 2 \cdot 2 = \boxed{16}$.

5. Since $1 + 1 = 2$ and $1 + 2 = 3$, the count for the 2 o'clock hour is the same as the count for the 11 o'clock hour, and the count for the 3 o'clock hour is the same as the count for the 12 o'clock hour. If the sum of the hour digits is 2, then the tens digit of the minute value can be any value from 0 through 4, for 5 possibilities. But if the sum of the hour digits is a 3, then the tens digit of the minute value can only be any value from 0 through 3, for 4 possibilities. Finally, if the sum of the hour digits is a 1, as it is in the 1 o'clock hour, then the tens digit of the minute value can be any value from 0 through 5, for 6 possibilities. Altogether, the number of minutes where the hour and minutes digits displayed sum to 6 is $5 + 4 + 6 + 5 + 4 = \boxed{24}$.

6. The area of the entire region of four squares is $4 \cdot 10 \cdot 10 = 400$. In the upper right of the region of four squares is an unshaded right triangle. This triangle has a width of 10 and is similar to the larger unshaded triangle that has a width of 40 and a height of 60. The height of the smaller triangle is $10 \cdot \frac{60}{40} = 15$, and the area of the smaller triangle is $\frac{1}{2} \cdot 10 \cdot 15 = 75$. In the upper left of the region of four squares is another unshaded right triangle. This triangle has a width of 10 and is similar to the larger unshaded triangle that has a width of 50 and height of 60. The height of the smaller triangle is $10 \cdot \frac{60}{50} = 12$, and the area of the smaller triangle is $\frac{1}{2} \cdot 10 \cdot 12 = 60$. The area of the portion of the shaded triangle that lies within the region of four squares is $400 - 75 - 60 = \boxed{265}$.

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7. The 3 identical squares have a side length that is 1 less than the square in the bottom middle of the rectangle. The square in the lower right of the rectangle has a side length that is 1 more than the side length of the square in the bottom middle. The square in the upper right has a side length that is both 1 more than the side length of the square in the lower right, and 1 less than 3 times the side length of the identical squares. Let the side length of the square in the bottom middle be x. The side length of the square in the upper right is $x + 2$ and the side length of the three identical squares is $x - 1$, so $3 \cdot (x - 1) - 1 = x + 2$. Solving, the side length of the middle bottom square is 3, so the side length of the three identical squares is 2, the side length of the lower right square is 4, and the side length of the large square is $3 + 2 + 2 + 2 = 9$. The width of the rectangle is $9 + 3 + 4 = 16$, and the height of the rectangle is 9, for an area of $16 \cdot 9 = \boxed{144}$.

8. Let the weight of Amy's bag, in pounds, be A and the weight of Bryan's bag, in pounds, be B. Additionally, let the weight limit be p and the cost per pound over the weight limit be c. Then $c \cdot (A - p) = 20$, and $c \cdot (B - p) = 10$. Additionally, $A + B = 120$, and $c \cdot (A + B - p) = 90$, or $cA + cB - cp = 90$. Summing the first two equations, $cA + cB - 2cp = 30$. Subtracting this from the last equation yields $cp = 60$, so $cA + cB = 90 + 60$. Since $A + B = 120$, the value of c is $\frac{90+60}{120} = 1.25$. Therefore $p = \frac{60}{1.25}$, so the weight limit, in pounds, is $\boxed{48}$.

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Team Round Solutions

1. The digit 4 appears 10 times in the units place, and an additional 10 times in the tens place. But 44 has the digit 4 twice, so the digit 4 occurs in $10 + 10 - 1 = 19$ numbers. Similarly, the digit 5 also occurs in 19 numbers, and the digit 6 also occurs in 19 numbers, for a total of $19 + 19 + 19 = 57$. However, 54, 45, 46, 64, 56, and 65 each occur in two counts, so the total of numbers that have the digit 4, the digit 5, or the digit 6 is $57 - 6 = \boxed{51}$.

2. The time between 5:45 and 8:45 is 3 hours, or $3 \cdot 60 = 180$ minutes. But 8:31 is $45 - 31 = 14$ minutes before 8:45, so the total time Travis spent watching the episodes was $180 - 14 = 166$ minutes. The first two episodes lasted a total of $47 + 50 = 97$ minutes, and the 2 breaks between the first and second episodes and the second and third episodes were $5 + 5 = 10$ minutes, for a total of $97 + 10 = 107$ minutes. Therefore the length of the third episode, in minutes, was $166 - 107 = \boxed{59}$.

3. No two-digit prime number can have a units digit of 5 or 4, so these two digits must be the tens digits of two of the three primes. To maximize the sum, the tens digit of the third number should be as large as possible. Therefore the third prime would have to be 97, as that is the only two-digit prime number with a tens digit of 9. The remaining two primes must be 41 and 53, so the greatest possible sum of the three primes is $97 + 41 + 53 = \boxed{191}$.

4. The diagram can be partitioned into a number of squares of side length $6 \div 2 = 3$. Of these these smaller squares, 7 are already visible in the diagram, and each of the two squares on either end of the diagram can be partitioned into 3 additional smaller squares. The total number of smaller squares is $7 + 3 + 3 = 13$. Each has an area of $3 \cdot 3 = 9$, so the total area of the region that lies within at least one of the squares is $9 \cdot 13 = \boxed{117}$.

5. Each of the cards numbered 5, 15, 25, 35, 45, and 55 must be in a separate group, as any two of those numbers will always sum to a multiple of 10. Place these cards into 6 groups. Each of the cards numbered 10, 20, 30, 40, and 50 must be placed into a different one of these 6 groups. The remaining cards can be sorted into those groups in any number of ways. One such way is to place the cards with a units digit of 1, 2, 3, or 4 in one group, and those with a units digit of 6, 7, 8, or 9 in another. Therefore the fewest number of groups required is $\boxed{6}$.

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6. The value of C is $\frac{A+B}{2}$, and the value of D is $\frac{1}{2} \cdot \left(B + \frac{A+B}{2}\right) = \frac{A+3B}{4}$. The value of 12 is $\frac{1}{2} \cdot \left(\frac{A+B}{2} + \frac{A+3B}{4}\right) = \frac{3A+5B}{8}$. Then $3A + 5B = 96$. If A is positive, then $3A$ must be both less than 96 and 1 more than a multiple of 5. The possible values of $3A$ are 6, 21, 36, 51, 66, and 81. But when $3A$ is 36, A is 12, and all other boxes have a value of 12, so this possibility must be discarded. Therefore the total number of possible values of A is $\boxed{5}$.

7. There are $\binom{9}{3} = 84$ possible sets of 3 dots. However, 3 dots will not form a triangle if the dots lie on the same line. Each of the 3 columns, each of the 3 rows, and each of the 2 diagonals have 3 dots on the same line. Therefore the total number of triangles that can be formed is $84 - 8 = \boxed{76}$.

8. Since each box is weighed with each of the 4 other boxes, each box is weighed four times. The total weight of all the weighings is $95+98+101+104+105+107+108+110+111+117 = 1056$ pounds, so the total weight of all five boxes is $\frac{1056}{4} = 264$. The two heaviest boxes must together weigh 117 pounds, and the two lightest boxes must together weigh 95 pounds. The weight of the third heaviest box must be $264 - 117 - 95 = 52$ pounds. The weight of the third heaviest box and the heaviest box must be 111 pounds, so the weight of the heaviest box must be $111 - 52 = 59$ pounds. The weight of the third heaviest box and lightest box must be 98 pounds, so the weight of the lightest box must be $98 - 52 = 46$ pounds, and the weight of the second lightest box must be $95 - 46 = 49$ pounds. The five boxes have weights of 46, 49, 52, 58, and 59 pounds. The product of the five weights is $46 \cdot 49 \cdot 52 \cdot 58 \cdot 59 = 401085776$, and the remainder when this value is divided by 100 is $\boxed{76}$.

9. The sum of the measures of the interior angles in a convex pentagon is $180^\circ \cdot (5-2) = 540^\circ$, so the measure of the interior angles at vertices C and D are each $\frac{1}{2} \cdot (540^\circ - 120^\circ - 90^\circ - 90^\circ) = 120^\circ$. The pentagon is symmetrical about a segment perpendicular to CD that passes through A, so side AB has length 18, and triangle ABE is isosceles. Let a segment from A that is perpendicular to BE meet BE at X. Triangle EAX is a $30-60-90$ triangle, as is triangle BAX, so the length of AX is $\frac{18}{2} = 9$, and the length of EX is $9\sqrt{3}$. The area of ABE is twice the area of EAX, or $2 \cdot \frac{1}{2} \cdot 9 \cdot 9\sqrt{3} = 81\sqrt{3}$. Quadrilateral $BCDE$ is an isosceles trapezoid, and segments CX and DX partition this trapezoid into three equilateral triangles, each with side length $9\sqrt{3}$. The total area of these three triangles is $3 \cdot \frac{(9\sqrt{3})^2 \cdot \sqrt{3}}{4} = \frac{729\sqrt{3}}{4}$. The total area of $ABCDE$ is $81\sqrt{3} + \frac{729\sqrt{3}}{4} = \frac{1053\sqrt{3}}{4}$, which when rounded to the nearest whole number is $\boxed{456}$.

10. There are $8 \cdot 8 \cdot 8 = 512$ possible outcomes when rolling the three dice. On each die, there are 3 sides that have a remainder of 1 when divided by 3, 3 sides that have a remainder of 2 when divided by 3, and 2 sides that have a remainder of 0 when divided by 3. The sum of the numbers shown will be divisible by three if all three dice show numbers that have the same remainder when divided by 3. The sum will also be divisible by 3 if each die shows a number that has a different remainder when divided by 3. There are $3 \cdot 3 \cdot 3 = 27$ outcomes where all three dice show a number with a remainder of 1, and an additional 27 outcomes where all three dice show a number with a remainder of 2. There are only $2 \cdot 2 \cdot 2 = 8$ outcomes where all three dice show a number with a remainder of 0. There are $3 \cdot 3 \cdot 2 = 18$ ways to select three numbers that have different remainders when divided by 3, and those three numbers can be permuted in $3! = 6$ ways among the three dice, for a total of $18 \cdot 6 = 108$ outcomes. Altogether, the total number of outcomes with sums that are divisible by 3 is $27 + 27 + 8 + 108 = 170$, for a probability of $\frac{170}{512} = \boxed{\frac{85}{256}}$.